Which invariant finer than L carries Smith's `full_success`?

`research/smith_class_invariant_20261015` · 2026-10-15 · protocol `PROTOCOL.md` frozen at `1461f9445`, Amendment 1 at `f6917f05b`, both before any null was run

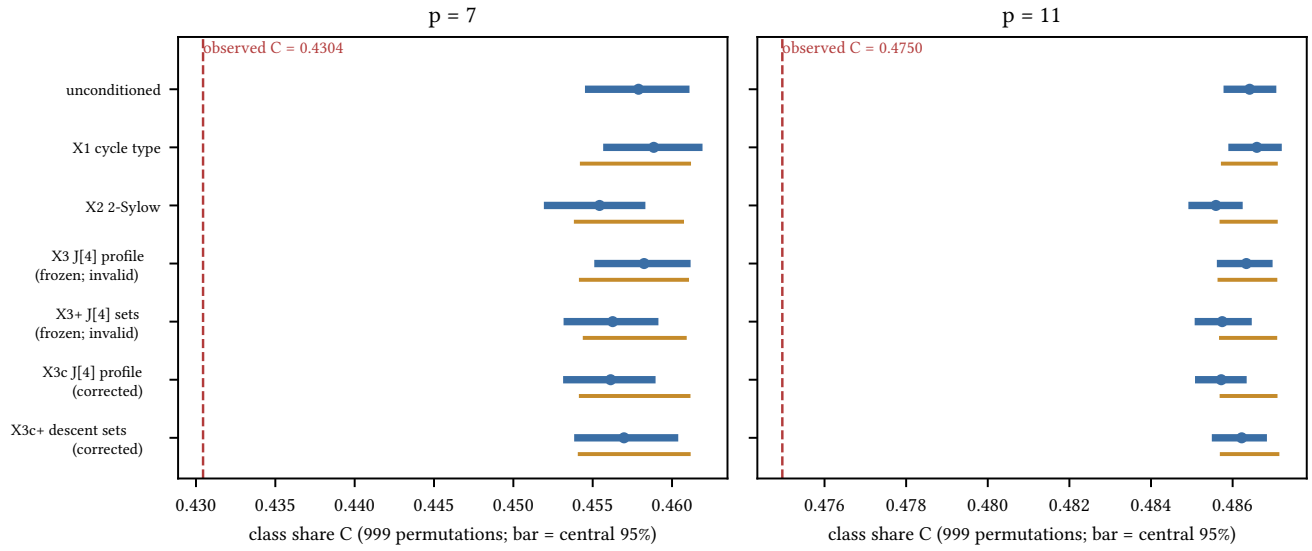
1. Question

At $p = 7$ and $p = 11$ Smith's genus-3 `full_success` clusters within L-polynomial isogeny classes beyond what independent coins per $\mathbb{F}_p$-isomorphism class give (`research/smith_iso_granularity_20261015`, observed class share C = 0.4304 / 0.475, below all 999 iso-level permutations). This study asks whether a Galois-module invariant finer than L (the cycle type of Frobenius on $J[2]$, the 2-Sylow subgroup of $J(\mathbb{F}_p)$, or Frobenius on $J[4]$) explains the clustering: conditioning the iso-level permutation null on it should bring C inside the null's central 95%, while an L-class placebo of equal cardinality should not.

2. Verdicts (frozen decision rule)

candidate	$p = 7$	$p = 11$	overall
X1: Frobenius cycle type on the Weierstrass points (finest $J[2]$ invariant)	0.4588 [0.4557, 0.4619]; gap share −0.035 **does not explain**	0.4866 [0.4859, 0.4872]; gap share −0.015 **does not explain**	**does not explain**
X2: 2-Sylow subgroup of $J(\mathbb{F}_p)$ (class group)	0.4554 [0.4519, 0.4583]; gap share 0.09 **does not explain**	0.4856 [0.4849, 0.4862]; gap share 0.072 **does not explain**	**does not explain**
X3: $\#J(\mathbb{F}_{p^k})[4]$, $k \mid 2m$ (frozen descent)	0.4582 [0.4551, 0.4612]; gap share −0.013 **invalid**	0.4863 [0.4856, 0.487]; gap share 0.007 **invalid**	**invalid**
X3+: halvable sets, canonical (frozen descent)	0.4563 [0.4532, 0.4591]; gap share 0.059 **invalid**	0.4857 [0.4851, 0.4865]; gap share 0.059 **invalid**	**invalid**
X3c: X3 corrected (Amendment 1)	0.4561 [0.4531, 0.459]; gap share 0.064 **does not explain**	0.4857 [0.4851, 0.4863]; gap share 0.061 **does not explain**	**does not explain**
X3c+: X3+ corrected (Amendment 1)	0.457 [0.4538, 0.4604]; gap share 0.033 **does not explain**	0.4862 [0.4855, 0.4868]; gap share 0.017 **does not explain**	**does not explain**

Cells: null median [2.5%, 97.5%] of C over 999 permutations within (stratum, invariant) cells; gap share $g_X = \frac{\text{med } N_0 - \text{med } N_X}{\text{med } N_0 - C}$ with N_0 the unconditioned iso-level null (median 0.4579 / 0.4864). "Invalid" is the frozen rule's outcome for X3/X3+, whose descent instrument failed control K4 (Section 4).



blue: null conditioned on (stratum, invariant), dot = median; orange: L-class placebo of equal cardinality. Source data/summary_p*.json.

Figure 1: Observed C (dashed) against each conditioned null (blue: central 95%, dot median) and its L-class placebo (orange), 999 permutations each, seed 20261015. Source `figures/plot.py`, data `data/summary_p*.json`.

3. Measured

quantity	p = 7	p = 11
tractable isomorphism classes (invariants computed on all)	16823	160010
observed C; unconditioned null 2.5–97.5% (K1 reproduction)	0.4304 0.4545–0.4611 (`true`)	0.475 0.4858–0.4871 (`true`)
cells; share of classes in single-class cells: X1_cycle_type	13 0	14 0
cells; share of classes in single-class cells: X2_sylow2	108 0.0008	184 0.0001
cells; share of classes in single-class cells: X3_j4_profile	32 0	37 0
cells; share of classes in single-class cells: X3plus_j4_sets	23 0.0001	26 0
cells; share of classes in single-class cells: X3c_j4_profile_corrected	26 0	28 0
cells; share of classes in single-class cells: X3cplus_j4_descent_sets_corrected	35 0.0002	44 0
X3c classes with an ambiguous $k > 1$ entry	3901	36681

4. Findings

- **Measured, frozen rule.** No candidate explains the within-class clustering. Conditioning the iso-level null on the cycle type (X1), the 2-Sylow subgroup of $J(\mathbb{F}_p)$ (X2) or the corrected $J[4]$ descent invariants (X3c, X3c+) leaves the observed C below every one of the 999 permutations at both primes. The share of the gap "unconditioned null median $-$ C" that conditioning removes is at most 9% (X2) and 6% (X3c); for X1 it is negative (−3.5% / −1.5%). The verdict for every valid candidate is **does not explain** at $p = 7$ and $p = 11$.
- **Measured.** The invariants do move the isomorphism-level success rate: within $\#S(H) = 1$ it ranges over 0.207–0.300 ($p = 7$) and 0.223–0.287 ($p = 11$) across cycle types, and 0.228–0.299 / 0.212–0.312 across 2-Sylow types with at least 200 classes (`descriptive_by_stratum_and_*` in the summaries). Rates that vary **between** cells but are shared by L-classes of every kind redistribute successes without concentrating them in particular L-classes, which is why the conditioned nulls barely move.
- **Measured, frozen rule.** X3 and X3+ (the frozen descent instrument) are **invalid**: control K4 failed on 4405 ($p = 7$) and 43538 ($p = 11$) classes, all of all-even cycle type, which Amendment 1 explains as the order-2 kernel of the even-degree descent map. Their nulls are recorded for completeness and are as far from C as the valid ones.
- **Derived, not a verdict.** X3c leaves 3901 / 36681 classes with an undecided $k > 1$ entry. Such classes are cells of their own value, so the invariant is, if anything, finer than the true $J[4]$ profile; a finer conditioning can only shrink the null toward C, and it did not.
- **Interpretation (labelled hypothesis).** The clustering is not carried by the 2-adic Galois module of J_H at the levels tested. A remaining structural candidate is the target of Smith's isogeny itself: `full_success` asks whether some $(2, 2, 2)$-quotient J_H/S with a rational trigonal map is a Jacobian over $\mathbb{F}_p$ rather than its quadratic twist (Smith Remark 4; Serre's obstruction). The quotients lie in the same L-class, so curves of one class share candidate quotients, which would correlate their outcomes. Testing this needs the isomorphism class of the quotient curve X per subgroup (e.g. Dixmier–Ohno invariants), not a further invariant of J_H.

5. Validation

control	p = 7	p = 11
K1: unconditioned null reproduces `summary_p*.json` of PR #1928 exactly	`true`	`true`
K2: X1/X3/X3c invariants equal on a second orbit member (every class)	pass	pass
K2: X2 equal on a second orbit member (1,000-class sample)	pass	pass
K3: every class certified; $\sum v_i = v_2(\#J(\mathbb{F}_p))$; rank $= \dim J(\mathbb{F}_p)[2]$	pass	pass
K4: frozen descent count at $k = 1$ equals class group (mismatching classes)	**fail** (4405)	**fail** (43538)
K5: frozen profile within the L-polynomial bounds (failing entries)	**fail** (9512)	**fail** (92783)
K4′: corrected $k = 1$ count brackets the class group	pass	pass
K5 for X3c	pass	pass
instrument errors (field membership, characters, relation labels)	0	0

K4 failed by design of the mathematics, not of the code: when every Frobenius orbit on the roots has even length, the even-degree descent map $J(K)/2J(K) \to L^*/K^*L^{*2}$ has a kernel of order exactly 2 (Amendment 1 gives the cohomological argument). The failures sit exactly on the all-even cycle types (8, 4.4, 2.6, 2.2.4, 4.2.2, 2.2.2.2), and the corrected count never disagrees with the class group (K4′).

6. Method

All computation is native Rust (`examples/smith_class_invariant.rs`, `compute` then `summarize`), on the representative model `y^2 = t f(x)` of every tractable isomorphism class of the frozen rows `iso_p{7,11}.csv.gz`.

- **X2.** Class group of the function field: generators all places of degree ≤ 3, relations $\mathrm{div}(a)$ for monic irreducible a and $\mathrm{div}(y - v)$ for random v of degree ≤ 4 whose norm $v^2 - tf$ is 3-smooth and squarefree; Smith form over $\mathbb{Z}/2^{e+1}$, accepted only when its order is 2^{2e+1}, which with $\mathrm{Pic} = \mathbb{Z} \oplus J(\mathbb{F}_p)$ certifies the 2-Sylow subgroup.
- **X3, X3c.** Roots of f in its splitting field $\mathbb{F}_{p^m}$ ($m \le 15$); descent images $\delta_{j(T_S)} = \prod_{i \in S, i \neq j} (e_i - e_j) \cdot f'(e_j)^{[j \in S]}$, and for swap-type points a rational representative from $y - \beta g_{S(x)}$ with $\beta^{q+1} = t$; quadratic characters by norms. X3c resolves the order-2 kernel with the Kummer pairing on the stable part and takes the $k = 1$ count from X2.
- **Nulls.** 999 permutations of `full` within (stratum, invariant) cells, seed 20261015; placebo: each L-class gets a hashed bucket among as many values as the invariant takes in the stratum.

7. Scope

Exact enumeration at $p = 7, 11$, degree-8 models only. X3c+ is an intrinsic invariant built from 2-descent data and is not claimed to be a complete invariant of Frobenius on $J[4]$; X3c leaves 4945 / 46779 profile entries as two-valued. A negative verdict excludes these invariants as explanations at these primes; it does not show that no 2-adic invariant matters. Nothing here builds or prices an isogeny walk or says anything at cryptographic sizes.

8. Reproduce

```
cargo build --release --example smith_class_invariant
B=target/release/examples/smith_class_invariant; D=research/smith_class_invariant_20261015/data
$B compute --p 7 --out $D; $B compute --p 11 --out $D     # 4 min and ~1 h on 2 vCPU
$B summarize --dir $D 7 11
python research/smith_class_invariant_20261015/figures/plot.py   # layout only
```

Run report: `reports/run-20261015T2332Z/report.pdf`. Prior: PR #1928, `research/smith_iso_granularity_20261015/`. Source: B. Smith, J. Cryptology 22 (2009), arXiv:0806.2995.